SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease)

Release Notes

This document provides guidance and an overview to high-level general features and updates for SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease). In addition to architecture- or product-specific information, it also describes the capabilities and limitations of SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease).

This product will be released in June 2025. The latest version of these release notes is always available at https://www.suse.com/releasenotes. Drafts of the general documentation can be found at https://susedoc.github.io/doc-slesforsap/main.

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1 About the release notes

These Release Notes are identical across all architectures, and the most recent version is always available online at https://www.suse.com/releasenotes.

Entries are only listed once but they can be referenced in several places if they are important and belong to more than one section.

Release notes usually only list changes that happened between two subsequent releases. Certain important entries from the release notes of previous product versions are repeated. To make these entries easier to identify, they contain a note to that effect.

However, repeated entries are provided as a courtesy only. Therefore, if you are skipping one or more service packs, check the release notes of the skipped service packs as well. If you are only reading the release notes of the current release, you could miss important changes.

2 SUSE Linux Enterprise Server for SAP Applications

SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) is the leading Linux platform for SAP HANA, SAP NetWeaver, and SAP S/4HANA applications that provides optimized performance, reduced downtime, and faster SAP landscape deployments. It includes features to secure SAP HANA systems and ease the transition to SAP S/4HANA for systems administrators. SUSE Linux Enterprise Server for SAP Applications combines SUSE Linux Enterprise Server and SUSE Linux Enterprise High Availability with additional software specifically meant to simplify running and managing SAP applications. This document provides an overview of high-level general features, capabilities and limitations of SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) and important product updates.

2.1 What Is New?

2.1.1 General Changes in SLE 15

Media Changes

The Unified Installer and Packages DVDs known from SUSE Linux Enterprise Server for SAP Applications 15 SP1 are deprecated and have been replaced by the following media:

- **Online Installation Media:** All SUSE Linux Enterprise 15 products can be installed with this stand-alone media, after entering a registration key. The necessary packages are fetched from online repositories only. For information about available modules, see https://www.suse.com/releasenotes/x86_64/SUSE-SLES/15-SP2#Intro.ModuleExtensionRelated

- **Full Installation Media:** All SUSE Linux Enterprise Server for SAP Applications 15 products can be installed without network connection with this media, for offline installation scenarios. The media contains all necessary packages. It consists of directories with module repositories which need to be added manually as needed. RMT (Repository Mirroring Tool) and SUSE Manager provide additional options for disconnected or managed installation.

Extended Package Search

Use the new Zypper command `zypper search-packages` to search across all SUSE repositories available for your product even if they are not yet enabled. This functionality makes it easier for administrators and system architects to find the software packages needed. To do so, it leverages the SUSE Customer Center.

Software Development Kit

With SLE 15, the Software Development Kit is now integrated into the products. Development packages are packaged alongside regular packages. Also, the *Development Tools* module contains the tools for development.

RMT Replaces SMT

SMT (Subscription Management Tool) has been removed. Instead, RMT (Repository Mirroring Tool) now allows mirroring SUSE repositories and custom repositories. You can then register systems directly with RMT. In environments with tightened security, RMT can also proxy other RMT servers.

MAJOR UPDATES TO THE SOFTWARE SELECTION:

Salt

SLE 15 SP7 (prerelease) can be managed via salt to help integration into up-to-date management solutions, such as SUSE Manager.

Python 3

As the first enterprise distribution, SLE 15 offers full support for Python 3 development.

Directory Server

389 Directory Server replaces OpenLDAP to provide a sustainable directory service.

2.1.2 General Changes in 15 SP7 (prerelease)

SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) introduces changes compared to SUSE Linux Enterprise Server for SAP Applications 15 SP6. The most important changes are listed below:

Kernel

SLE 15 SP7 (prerelease) comes with Kernel 6.4. This new kernel release includes upstream features like 16 million new IPv4 addresses, utilization clamping support in the task scheduler, power-efficient userspace waiting with the umwait x86 instructions and many more.

2.2 Important Sections of This Document

If you are upgrading from a previous SUSE Linux Enterprise Server for SAP Applications release, you should review at least the following sections:

- *Section 2.6, "Support statement for SUSE Linux Enterprise Server for SAP Applications"*

2.3 Security, Standards, and Certification

SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) does not have its own NIST FIPS 140-3 certification. Instead, it uses vendor affirmation for these cryptographic modules that SUSE certified for SUSE Linux Enterprise Server 15 SP6: OpenSSL 3, OpenSSL 1, GnuTLS, Libgcrypt, Libica, and NSS. This does not include the kernel. The certified packages are in the Certifications Module.

For more information about certification, see https://www.suse.com/security/certificates.html.

2.4 Documentation and other information

2.4.1 Available on the product media

- Read the READMEs on the media.
- Get the detailed change log information about a particular package from the RPM (where *FILENAME*.rpm is the name of the RPM):

```
rpm --changelog -qp FILENAME.rpm
```

- Check the ChangeLog file in the top level of the installation medium for a chronological log of all changes made to the updated packages.
- Find more information in the docu directory of the installation medium of SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease). This directory includes PDF versions of the SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) Installation Quick Start Guide.
- Get list of manual pages with usage information about a particular package from the RPM (where *FILENAME*.rpm is the name of the RPM):

```
rpm --docfiles -qp FILENAME.rpm | grep man
```

2.4.2 Online documentation

- For the most up-to-date version of the documentation for SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease), see https://susedoc.github.io/doc-slesforsap/main (draft version).

2.5 Support and lifecycle

SUSE Linux Enterprise Server for SAP Applications is backed by award-winning support from SUSE, an established technology leader with a proven history of delivering enterprise-quality support services.

SUSE Linux Enterprise Server for SAP Applications includes Extended Service Pack Overlay Support (ESPOS). This provides an additional 3 years of support for each SP following the end of general support. Combined with the standard 6 months overlay, it offers a total of 3.5 years of overlay support after the launch of the next Service Pack.

For more information, check our Support Policy page https://www.suse.com/support/policy.html or the Long Term Service Pack Support Page https://www.suse.com/support/programs/long-term-service-pack-support.html.

2.5.1 Quarterly updates

SUSE provides quarterly updates for SUSE Linux Enterprise Server for SAP Applications 15 SP7 (pre-release) until the end of General Support on 31 Jul 2031. SUSE refreshes the installation media, virtual machine (VM) images, and public cloud images at least quarterly.

2.6 Support statement for SUSE Linux Enterprise Server for SAP Applications

To receive support, you need an appropriate subscription with SUSE. For more information, see https://www.suse.com/support/programs/subscriptions/?id=SUSE_Linux_Enterprise_Server_for_SAP_Applications.

The following definitions apply:

L1

Problem determination, which means technical support designed to provide compatibility information, usage support, ongoing maintenance, information gathering and basic troubleshooting using available documentation.

L2

Problem isolation, which means technical support designed to analyze data, reproduce customer problems, isolate problem area and provide a resolution for problems not resolved by Level 1 or prepare for Level 3.

L3

Problem resolution, which means technical support designed to resolve problems by engaging engineering to resolve product defects which have been identified by Level 2 Support.

For contracted customers and partners, SUSE Linux Enterprise Server for SAP Applications is delivered with L3 support for all packages, except for the following:

- Technology Previews, see *Section 2.7, "Technology Previews"*
- Sound, graphics, fonts and artwork

- Certain packages shipped as part of the module *Workstation Extension* are L2-supported only
- Packages with names ending in `-devel` (containing header files and similar developer resources) are only supported together with their main packages.

SUSE only supports the usage of original packages. That is, packages that are unchanged and not recompiled.

2.6.1 General support

To learn about supported features and limitations, refer to the following sections in this document:

- *Section 5, "New Features, Changes & Fixes"*
- *Section 6, "Known Issues & Workarounds"*

2.6.2 Software requiring specific contracts

Certain software delivered as part of SUSE Linux Enterprise Server for SAP Applications may require an external contract. Check the support status of individual packages using the RPM metadata that can be viewed with `rpm`, `zypper`, or YaST.

Major packages and groups of packages affected by this are:

- PostgreSQL (all versions, including all subpackages)
- `sap-convergent-resource-agents`

2.6.3 SAP Support Statement

Make sure to also review the SAP Product Availability Matrix (https://apps.support.sap.com/sap/support/pam) and/or SAP Note 2235581 - SAP HANA: Supported Operating Systems (https://launchpad.support.sap.com/#/notes/2235581) for official SAP support and certification statements.

2.6.4 Software under GNU AGPL

SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) (and the SUSE Linux Enterprise modules) includes the following software that is shipped *only* under a GNU AGPL software license:

- Ghostscript (including subpackages)

SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) (and the SUSE Linux Enterprise modules) includes the following software that is shipped under multiple licenses that include a GNU AGPL software license:

- MySpell dictionaries and LightProof
- ArgyllCMS

2.7 Technology Previews

Technology previews are packages, stacks or features delivered by SUSE which are not supported. They may be functionally incomplete, unstable or in other ways not suitable for production use. They are included for your convenience and give you a chance to test new technologies within an enterprise environment.

Whether a technology preview becomes a fully supported technology later depends on customer and market feedback. Technology previews can be dropped at any time and SUSE does not commit to providing a supported version of such technologies in the future.

Give your SUSE representative feedback about technology previews, including your experience and use case.

3 Modules, Extensions, and Related Products

This section comprises information about modules and extensions for SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease). Modules and extensions add functionality to the system.

3.1 Modules in the SLE 15 SP7 (prerelease) Product Line

The SLE 15 SP7 (prerelease) product line is made up of modules that contain software packages. Each module has a defined scope. Modules differ in their life cycles and update timelines.

The modules available within the product line based on SUSE Linux Enterprise 15 SP7 (prerelease) at the release of SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) are listed in the *Modules and Extensions Quick Start* at https://susedoc.github.io/doc-sle/main/html/SLES-modulesquick/ (draft version).

Not all SLE modules are available with a subscription for SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) itself (see the column *Available for*).

For information about the availability of individual packages within modules, see https://scc.suse.com/packages.

3.2 Available Extensions

Extensions add extra functionality to the system and require their own registration key, usually at additional cost. Most extensions have their own release notes documents that are available from https://www.suse.com/releasenotes.

The following extensions are available for SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease):

- SUSE Linux Enterprise Live Patching: https://www.suse.com/products/live-patching

The following extension is not covered by SUSE support agreements, available at no additional cost, and without an extra registration key:

- SUSE Package Hub: https://packagehub.suse.com/

3.3 Derived and Related Products

This sections lists derived and related products. Usually, these products have their own release notes documents that are available from https://www.suse.com/releasenotes.

- SUSE Linux Enterprise Server: https://www.suse.com/products/server
- SUSE Linux Enterprise Desktop: https://www.suse.com/products/desktop
- SUSE Linux Enterprise for High-Performance Computing: https://www.suse.com/products/server/hpc
- SUSE Linux Enterprise Real Time: https://www.suse.com/products/realtime
- SUSE Manager: https://www.suse.com/products/suse-manager

4 Changes affecting all architectures

Information in this section applies to all architectures supported by SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease).

5 New Features, Changes & Fixes

Information in this section applies to all architectures supported by SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) unless noted otherwise.

5.1 HA for SAP Convergent Mediation

The package `sap-convergent-resource-agents` ships resource agent (RA) for making the SAP Convergent Mediation (CM) ControlZone (by Digital Route) highly available. The RA uses CM's `mzsh` as interface. See man page `ocd_suse_SAPCMControlZone(7)` for details.

This package is supported only in the context of SAP RISE.

5.2 Silence `KillMode=None` messages

The log level of the deprecation warnings regarding `killmode=None` have been reduced. Instead of `warning`, they are now logged at the `debug` log level.

5.3 Enriched system visibility in the SUSE Customer Center (SCC)

SUSE is committed to helping provide better insights into the consumption of SUSE subscriptions regardless of where they are running or how they are managed; physical or virtual, on-premises or in the cloud, connected to SCC or Repository Mirroring Tool (RMT), or managed by SUSE Manager. To help you identify or filter out systems in SCC that are no longer running or decommissioned, SUSEConnect now features a daily "ping", which updates system information automatically.

For more details see the documentation at https://documentation.suse.com/subscription/suseconnect/single-html/SLE-suseconnect-visibility/.

5.4 SAPHanaSR-angi: SAP HANA SR - Advanced Next Generation Interface

SAPHanaSR-angi is "SAP HANA SR - Advanced Next Generation Interface" for SUSE high availabilty clusters to manage SAP HANA databases with system replication. Goal of SAPHanaSR-angi is consolidating and finally replacing the SAPHanaSR and SAPHanaSR-ScaleOut packages. Nevertheless, CIB attributes are not backward compatible between SAPHanaSR-angi and SAPHanaSR or SAPHanaSR-ScaleOut. Upgrading from SAPHanaSR or SAPHanaSR-ScaleOut requires a certain procedure, see man page SAPHanaSR_upgrade_to_angi(7) for details. New deployments should be done with SAPHanaSR-angi, see man page SAPHanaSR-angi(7) for further information.

6 Known Issues & Workarounds

This is a list of known issues for this release.

6.1 Workarounds to Enable SAP HANA Workloads on Intel Optane DC Memory

SUSE Linux Enterprise Server for SAP Applications 15 supports Intel Optane DC persistent memory. This enables SAP workloads, such as SAP HANA to benefit from persistent memory This shortens start times of the system and provides better system stability.

From a file system perspective, the XFS file system is supported for persistent memory, with SAP HANA running in DAX mode. Make sure to also check the SAP instructions for using persistent memory with SAP HANA. SUSE is working closely with SAP on future developments.

If there are pmem namespaces, these need to be destroyed before the installation. To mount persistent memory directly on boot, the respective entries need to be added to /etc/fstab, for example:

```
/dev/pmem0    /mnt/pmem0    xfs    dax    0  0
/dev/pmem1    /mnt/pmem1    xfs    dax    0  0
```

Namespaces need to be created individually. Execute the following command for each namespace you want to create:

```
ndctl create-namespace --mode=fsdax --map=dev
```

Because the systemd and udev default timeout may be too short for large memory setups to become usable, your system may run into a timeout and the boot process may stop with an emergency shell. To prevent this from happening, increase the timeout to 1200 seconds. Create the file `/etc/systemd/system/systemd-udev-settle.service.d/00-override.conf` with the following content:

```
[Service]
TimeoutSec=1200s
ExecStart=/usr/bin/udevadm settle -t 300
```



Important

In an earlier version of this document, we recommended mounting persistent memory with the `nofail` option in `/etc/fstab`. This frequently required a manual mount of pmem devices on reboot. To automatically mount pmem file systems after reboot, we strongly recommend removing the `nofail` option and modifying the timeout value as described above.

7 Removed and Deprecated Features and Packages

This section lists features and packages that got removed from SUSE Linux Enterprise Server for SAP Applications or will be removed in upcoming versions.

7.1 Deprecated Features and Packages

The following features and packages are deprecated and will be removed in the future.

- `ocfs2` has been deprecated.

7.1.1 SAPHanaSR, SAPHanaSR-ScaleOut and filesystem resource agent deprecations

We are announcing some important changes to HA cluster resource agents and packages in SLES for SAP 16.0. Please be aware of the following deprecations and plan your deployments accordingly:

Feature	Deprecated Component	Replacement Component	Supported in SLES for SAP 15 SP7 (prerelease)	Supported in SLES for SAP 16.0
HA for SAP HANA database	`SAPHanaSR`, `SAPHanaSR-ScaleOut`	`SAPHanaSR-angi`	Yes (but migration recommended)	Yes
HA for SAP S/4HANA & NetWeaver Central Services	Filesystem resource agent, NetWeaver	simple-mount solution with `SAPStartSrv` resource agent	Yes (but migration recommended)	No

1. HA for SAP HANA database

    - The `SAPHanaSR` and `SAPHanaSR-ScaleOut` packages are deprecated and will be replaced by `SAPHanaSR-angi` in SLES for SAP 16.0.
    - We recommend using `SAPHanaSR-angi` for new deployments, even on SLES for SAP 15 SP7 (prerelease). This new package offers improved performance and scalability.
    - Existing clusters using `SAPHanaSR` or `SAPHanaSR-ScaleOut` on SLES for SAP 15 SP7 (prerelease) will remain supported. For more information on `SAPHanaSR-angi` see

- SAP HANA System Replication Scale-Up - Performance Optimized Scenario (https://documentation.suse.com/sbp/sap-15/html/SLES4SAP-hana-an-gi-perfopt-15/index.html)
- SAP HANA System Replication Scale-Out - Performance-Optimized Scenario (https://documentation.suse.com/sbp/sap-15/html/SLES4SAP-hana-an-gi-scaleout-perfopt-15/index.html)

2. HA for SAP S/4HANA and SAP NetWeaver Central Services

   - The HA solution based on the Filesystem resource agent is deprecated and will not be supported in SLES for SAP 16.0.
   - Instead, use the simple-mount solution based on the `SAPStartSrv` resource agent. This is the recommended approach for new deployments on SLES for SAP 15 SP7 (prerelease) and later.
   - Existing clusters using the Filesystem resource agent on SLES for SAP 15 SP7 (prerelease) will remain supported.
   - The HA solution for the SAP NetWeaver central services agent is deprecated and will not be supported in SLES for SAP 16.0.
   - Instead, use the S/4HANA central services. This is the recommended approach for new deployments on SLES for SAP 15 SP7 (prerelease) and later.
   - Existing clusters using SAP NetWeaver on SLES for SAP 15 SP7 (prerelease) will remain supported.

For detailed information on setting up the simple-mount solution, see SAP S/4 HANA - Enqueue Replication 2 High Availability Cluster With Simple Mount (https://documentation.suse.com/sbp/sap-15/html/SAP-S4HA10-setupguide-simplemount-sle15/index.html)

These changes are being made to simplify deployment, improve performance, and enhance the overall HA experience in SLES for SAP.

7.1.2 Additional legacy tool deprecated

The following legacy tools are deprecated and will be removed in a future package version:

- `SAPHanaSR-filter-legacy`
- `SAPHanaSR-monitor-legacy`
- `SAPHanaSR-replay-archive-legacy`
- `SAPHanaSR-showAttr-legacy`

These tools are not compatible with the new `SAPHanaSR-angi` package. Customers using `SAPHanaSR-angi` should transition away from these legacy tools.

8 Obtaining source code

This SUSE product includes materials licensed to SUSE under the GNU General Public License (GPL). The GPL requires SUSE to provide the source code that corresponds to the GPL-licensed material. The source code is available for download at https://www.suse.com/download/sle-sap/ on Medium 2. For up to three years after distribution of the SUSE product, upon request, SUSE will mail a copy of the source code. Send requests by e-mail to sle_source_request@suse.com (mailto:sle_source_request@suse.com). SUSE may charge a reasonable fee to recover distribution costs.

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10 Changelog for 15 SP7 (prerelease)

10.1 Pre-release

10.1.1 Updated

- Aligned the FIPS certification status in *Section 2.3, "Security, Standards, and Certification"* with SUSE Linux Enterprise Server (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1289040) ↗)

10.2 2026-10-01

10.2.1 New

- Added note about quarterly updates for SUSE Linux Enterprise Server for SAP Applications 15 SP7 (prerelease) to *Section 2.5.1, "Quarterly updates"* (Jira (https://jira.suse.com/browse/PED-14584) ↗).

10.2.2 Updated

- Converted changelog appendix to regular section to improve navigation in single-page HTML.
- Updated the wording of *Section 7.1.1, "`SAPHanaSR`, `SAPHanaSR-ScaleOut` and filesystem resource agent deprecations"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1274567) ↗)
- Updated the Python 3 statement to exclude Python 2 (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1205940) ↗)
- Changed documentation links to point to the SP7 documentation (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1258177) ↗)

10.3 2025-06-16

10.3.1 New

- Added `ocfs2` deprecation note to *Section 7.1, "Deprecated Features and Packages"* (Jira (https://jira.suse.com/browse/PED-11048))
- *Section 7.1.1, "`SAPHanaSR`, `SAPHanaSR-ScaleOut` and filesystem resource agent deprecations"* (Jira (https://jira.suse.com/browse/PED-11566))

10.4 2024-09-12

10.4.1 New

- Initial 15 SP7 (prerelease) release